The Portfolio: Work Continues

NoteMates: Your Study Buddy for Your Notes
Part 2 (Draft): Features, Resources and User Walkthrough

Features

Version 1 (Part 1) lets you add notes and ask questions. Version 2 builds on that with:

1. **Notes upload:** add your notes as a document or paste the text.
2. **Study chatbot:** ask questions in plain English and get answers based on your notes.
3. **Quiz generator:** automatically creates multiple-choice practice questions from your notes, in the style of the Cloud Practitioner exam.
4. **Flashcards:** pulls out key terms and definitions for quick revision.
5. **Topic summary:** condenses a long section of notes into key points.
6. **Beginner mode:** re-explains a topic in simple, non-technical language.

Resources required

Prototype (PartyRock):

- **Foundation model:** PartyRock runs on Amazon Bedrock, so no model setup is needed.
- **Widgets:** a document upload widget, a text input, a chatbot widget, and text generation widgets for the quiz, flashcards, and summary, each referencing the notes.
- **Storage:** no database. Notes are supplied per app session, which keeps v1 simple. The trade-off is that you re-add your notes each time and NoteMates doesn't remember your progress.

Production version (what I'd use to take NoteMates further):

- **Amazon S3:** stores uploaded notes.
- **Amazon Bedrock Knowledge Bases:** indexes the notes so the chatbot retrieves relevant sections (retrieval-augmented generation, or RAG).
- **Vector store (e.g., Amazon OpenSearch Serverless):** holds the embeddings the Knowledge Base searches.
- **Amazon DynamoDB:** stores user profiles, quiz scores, and chat history.
- **Amazon Cognito:** handles sign-in so each student's notes stay private.
- **AWS IAM:** least-privilege permissions between services.
- **Bedrock Guardrails:** keeps responses on topic and filters unwanted content.

User walkthrough

1. You open NoteMates and see a welcome message and an upload box.
2. You upload your notes, or paste them in.
3. You ask, "What's the difference between S3 Standard and S3 Glacier?" NoteMates answers using your notes.

4. You ask it to explain the answer more simply, and beginner mode rewrites it in plain language.
5. You click to generate a quiz on storage services and answer the questions to test yourself.
6. You review the flashcards for terms you missed, then ask for a summary of the section before moving on.

Flow diagram

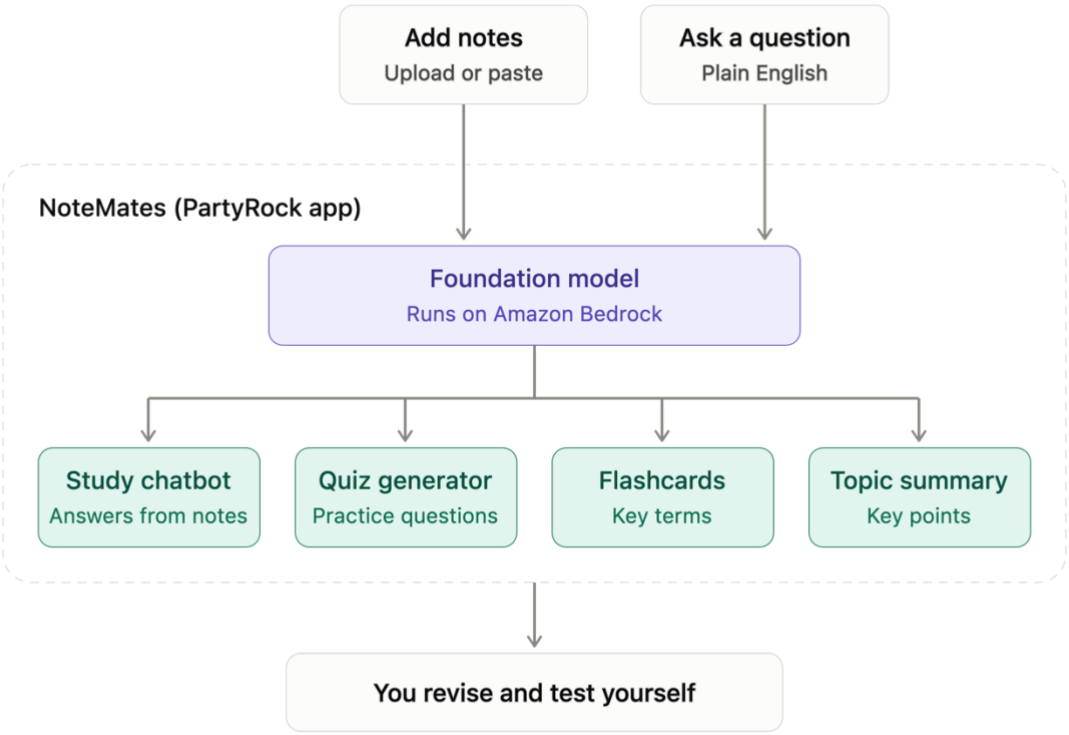


Figure 1: How NoteMates turns a student's notes and questions into answers, quizzes, flashcards and summaries.

Responsible AI note

Because the model can occasionally get things wrong, NoteMates should answer from your notes and say so when the answer isn't in them, rather than guess. You should double-check anything important against official AWS documentation.